Homework 1

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Question 1: A First Look at Denoising Models

(a) Posterior. Both $p(x_0)$ and $p(x_1 \mid x_0)$ are Gaussian with mean linear in x_0, so (x_0, x_1) is jointly Gaussian and the posterior is Gaussian before any algebra; the work is to find its two parameters. By Bayes' rule, keeping only the factors that involve x_0,

$$p(x_0 \mid x_1) \;\propto\; p(x_0)\, p(x_1 \mid x_0) \;\propto\; \exp\left\{-\frac{x_0^2}{2\tau_0^2} - \frac{(x_1 - \sqrt{1-\beta}\, x_0)^2}{2\beta}\right\}.$$

Expanding the square and dropping $x_1^2/(2\beta)$, which is constant in x_0, the exponent becomes

$$-\frac{x_0^2}{2\tau_0^2} - \frac{(1-\beta)\, x_0^2 - 2\sqrt{1-\beta}\, x_0 x_1}{2\beta} \;=\; -\frac{1}{2}\underbrace{\left(\frac{1}{\tau_0^2} + \frac{1-\beta}{\beta}\right)}_{\lambda} x_0^2 \;+\; \underbrace{\frac{\sqrt{1-\beta}\, x_1}{\beta}}_{h}\, x_0.$$

Completing the square, any density proportional to $\exp\{-\frac{1}{2}\lambda x_0^2 + h x_0\}$ is $\mathcal{N}(h/\lambda,\, 1/\lambda)$. Here $\lambda = \big(\beta + (1-\beta)\tau_0^2\big)/(\beta\tau_0^2)$, so

$$\frac{1}{\lambda} = \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}, \qquad \frac{h}{\lambda} = \frac{\sqrt{1-\beta}\, x_1}{\beta} \cdot \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2} = \frac{\sqrt{1-\beta}\,\tau_0^2\, x_1}{\beta + (1-\beta)\tau_0^2},$$

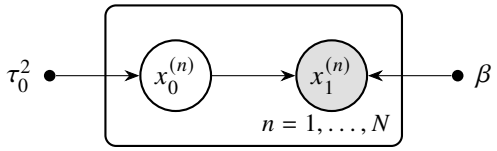
$$\boxed{x_0 \mid x_1 \;\sim\; \mathcal{N}\!\left(\frac{\sqrt{1-\beta}\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}\, x_1,\;\; \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}\right).}$$

Check. Jointly $\mathbb{E}[x_1] = 0$, $\mathrm{Var}(x_1) = (1-\beta)\tau_0^2 + \beta$ by total variance, and $\mathrm{Cov}(x_0, x_1) = \mathbb{E}[x_0\, \mathbb{E}(x_1 \mid x_0)] = \sqrt{1-\beta}\,\tau_0^2$; Gaussian conditioning, mean $\mathrm{Cov}(x_0, x_1)\, x_1/\mathrm{Var}(x_1)$ and variance $\tau_0^2 - \mathrm{Cov}(x_0, x_1)^2/\mathrm{Var}(x_1)$, reproduces both boxed expressions.

The mean is $x_1/\sqrt{1-\beta}$, the inversion of the deterministic scaling, times $(1-\beta)\tau_0^2/\big((1-\beta)\tau_0^2 + \beta\big)$, the share of $\mathrm{Var}(x_1)$ due to signal rather than injected noise; the posterior therefore shrinks that inversion toward the prior mean 0. Its variance is free of x_1: an observation relocates the posterior without changing its spread.

(b) Limits. As $\beta \to 0$ the mean tends to x_1 and the variance to 0: the step is the identity, so x_1 pins down x_0 exactly and the posterior is a point mass. As $\beta \to 1$ the mean tends to 0 and the variance to τ_0^2: x_1 is pure noise, the likelihood is independent of x_0, and the posterior falls back on the prior $\mathcal{N}(0, \tau_0^2)$. Since $d\sigma^2/d\beta = \tau_0^4/\big(\beta + (1-\beta)\tau_0^2\big)^2 > 0$, the variance climbs monotonically between these limits, so β sets how much of x_0 survives the step.

(c) Graphical model. The statement describes one (x_0, x_1) pair; the plate shows i.i.d. replication over N pairs (take $N = 1$ here). Shaded $x_1^{(n)}$ is observed, $x_0^{(n)}$ is latent, and the fixed τ_0^2, β sit outside the plate, shared across all n.



Question 2: Discrete Conjugacy

(a) Posterior. Write $\alpha_0 = \sum_{v=1}^{V} \alpha_v$ and $B(\boldsymbol{a}) = \prod_{v=1}^{V} \Gamma(a_v)/\Gamma\big(\sum_{v=1}^{V} a_v\big)$, so the prior density on the simplex $\Delta_{V-1} = \{\boldsymbol{\pi} : \pi_v \geq 0, \sum_v \pi_v = 1\}$ is $p(\boldsymbol{\pi}) = B(\boldsymbol{\alpha})^{-1} \prod_{v=1}^{V} \pi_v^{\alpha_v - 1}$; equivalently $\int_{\Delta_{V-1}} \prod_{v=1}^{V} \pi_v^{a_v - 1}\, d\boldsymbol{\pi} = B(\boldsymbol{a})$ for any $\boldsymbol{a}$ with positive entries, an identity reused in part (b). The likelihood is where the counts appear. Writing one categorical draw as $p(y_i \mid \boldsymbol{\pi}) = \pi_{y_i} = \prod_{v=1}^{V} \pi_v^{\mathbf{1}\{y_i = v\}}$ and using conditional independence given $\boldsymbol{\pi}$,

$$p(y_1, \ldots, y_n \mid \boldsymbol{\pi}) = \prod_{i=1}^{n} \prod_{v=1}^{V} \pi_v^{\mathbf{1}\{y_i = v\}} = \prod_{v=1}^{V} \pi_v^{\sum_{i=1}^{n} \mathbf{1}\{y_i = v\}} = \prod_{v=1}^{V} \pi_v^{n_v},$$

so the counts $(n_1, \ldots, n_V)$ are sufficient. The likelihood has the prior's product-of-powers form; multiplying adds the exponents:

$$p(\boldsymbol{\pi} \mid y_1, \ldots, y_n) \;\propto\; \prod_{v=1}^{V} \pi_v^{\alpha_v - 1} \prod_{v=1}^{V} \pi_v^{n_v} \;=\; \prod_{v=1}^{V} \pi_v^{(\alpha_v + n_v) - 1}.$$

The right side is the Dirichlet kernel with parameters $\alpha_v + n_v > 0$; a density is pinned down by its kernel, so the normalizer is $B(\boldsymbol{\alpha} + \boldsymbol{n})^{-1}$ and

$$\boxed{\boldsymbol{\pi} \mid y_1, \ldots, y_n \;\sim\; \text{Dirichlet}(\alpha_1 + n_1,\; \ldots,\; \alpha_V + n_V).}$$

The prior parameters behave as pseudocounts: the data simply add n_v to α_v, and the concentration grows from α_0 to $\alpha_0 + n$.

(b) Posterior predictive. Let $\boldsymbol{\alpha}' = \boldsymbol{\alpha} + \boldsymbol{n}$, so $\alpha'_0 = \sum_u \alpha'_u = \alpha_0 + n$. Given $\boldsymbol{\pi}$ the next draw is independent of the previous ones, so conditioning on $\boldsymbol{\pi}$ and marginalizing it out,

$$p(y_{n+1} = v \mid y_{1:n}) = \int_{\Delta_{V-1}} \pi_v \, p(\boldsymbol{\pi} \mid y_{1:n})\, d\boldsymbol{\pi} = \mathbb{E}[\pi_v \mid y_{1:n}],$$

The factor π_v increments the v-th exponent, so the beta-function identity gives

$$\begin{aligned} \mathbb{E}[\pi_v \mid y_{1:n}] &= \frac{1}{B(\boldsymbol{\alpha}')} \int_{\Delta_{V-1}} \prod_{u=1}^{V} \pi_u^{(\alpha'_u + \mathbf{1}\{u=v\}) - 1}\, d\boldsymbol{\pi} = \frac{B(\boldsymbol{\alpha}' + \boldsymbol{e}_v)}{B(\boldsymbol{\alpha}')} \\ &= \frac{\Gamma(\alpha'_v + 1)}{\Gamma(\alpha'_v)} \cdot \frac{\Gamma(\alpha'_0)}{\Gamma(\alpha'_0 + 1)} = \frac{\alpha'_v}{\alpha'_0}. \end{aligned}$$

The $u \neq v$ factors cancel and $\Gamma(z+1) = z\,\Gamma(z)$ is used twice. Hence

$$\boxed{p(y_{n+1} = v \mid y_1, \ldots, y_n) = \frac{\alpha_v + n_v}{\alpha_0 + n}, \qquad \alpha_0 = \sum_{u=1}^{V} \alpha_u.}$$

The probabilities sum to one since $\sum_v (\alpha_v + n_v) = \alpha_0 + n$. For $n > 0$, the predictive is $\frac{n}{\alpha_0 + n} \frac{n_v}{n} + \frac{\alpha_0}{\alpha_0 + n} \frac{\alpha_v}{\alpha_0}$, a convex combination of the MLE and prior mean. Thus α_0 is an effective prior sample size: prediction tends to the MLE as $n \to \infty$ and remains positive for unseen categories. This is additive smoothing; $\alpha_v = 1$ gives Laplace smoothing.

Question 3: Box's Loop

(1) Question. Among the available Boston Marathon race records, does a two-component Gaussian mixture predict late-race slowing better than a single Gaussian or a single skew-normal, and would that justify calling its components "wall" and "no wall"? These are separate questions: a useful density approximation need not identify two physiological states.

(2) Data and processing. I use the public llimllib/bostonmarathon scrape of BAA results. Its scraper retrieves individual race-result pages and exports cumulative checkpoint times in minutes, rounded to two decimals. The 2013 and 2014 files contain 16,164 and 31,984 rows; variables include bib, recorded sex, age, and times at 5, 10, 20, 21.0975, 25, 30, 35, 40 and 42.195 km. A row is a race performance, not necessarily a distinct person across years; the exported files omit the 15 km split.

I remove 105 wheelchair records, 7 repeated bibs within a year, and 41 records with an available segment pace outside the screening range 2–20 min/km. Missing tokens and nonpositive times become missing; non-increasing cumulative times would also be blanked (0 occur here). The loader interpolates isolated interior gaps, but the analysis requires all four defining endpoints to be observed, excluding another 304 records: 47,691 remain, 457 fewer than the raw files. A missing split's cause is unknown, and the slow-pace cutoff can remove a real stop, so these are selection decisions rather than proof of faulty timing.

For cumulative time T_d at distance d, define

$$a_i = \frac{T_{25,i} - T_{10,i}}{15}, \qquad r_i = \log\left(\frac{(T_{40,i} - T_{30,i})/10}{a_i}\right), \qquad y_i = r_i - \widehat{q}_{.25}(a_i).$$

Here a_i is baseline pace and $r_i > 0$ means the 30–40 km window was slower. Negative y_i means less slowing than the reference, not necessarily a faster late pace. The reference is the training sample's 25th percentile of r in 40 baseline-pace bins, interpolated between bin medians and held constant beyond them. Figure 1a explores the raw relationship; panels b–c show the centered distributions. This is empirical centering, *not* a terrain correction: the 30–40 km window still includes Heartbreak Hill (BAA course map). The pooled rise near 5.5–6 min/km is consistent with 2013 interruption-related finish-time truncation and changing year composition; fitting both the reference and models on 2014 training records alone gives 169.8 held-out nats over the skew-normal (s.e. 41.4).

(3) Scope and limitations, set before the refit. These records can assess density fit in the observed sample, but contain no direct observation of glycogen depletion, fueling, pain or perceived exhaustion. The source excludes nonfinishers and acknowledges missing finishers; the 2013 race interruption adds severe selection, so neither a population wall rate nor a lower bound on it is identified. Boston's entry rules and course further limit generalization. Nonfinisher splits and independently collected fueling, heart-rate and exhaustion records would help distinguish timing patterns from a physiological event.

(4) Model and fitting. For each recorded-sex × year group g, I fit $\mathcal{M}_0 : \; y_i \sim \mathcal{N}(\mu_g, \sigma_g^2)$, a single skew-normal $\mathcal{M}_s$, and the latent-class model

$$z_i \sim \text{Bernoulli}(\pi_g), \qquad y_i \mid z_i = k \sim \mathcal{N}(\mu_{k,g}, \sigma_{k,g}^2), \quad \mu_{0,g} < \mu_{1,g}.$$

I split records 80/20 within groups (seed 20261016; 38,152 training, 9,539 held out), estimate the reference only on training records, and fit by maximum likelihood: closed form for $\mathcal{M}_0$, six-start numerical optimization for $\mathcal{M}_s$, and six-start EM for the mixture, alternating probabilistic component assignments with weighted parameter updates. To prevent the mixture's unbounded likelihood at a collapsed component, each standard deviation is constrained to at least $\max(10^{-4}, 0.02 s_g)$, where s_g is the training standard deviation; all selected fits converge and none uses that floor.

(5) Results and criticism. For each held-out record I compute $d_i = \log \widehat{p}_2(y_i) - \log \widehat{p}_s(y_i)$; positive values favor the mixture. The table reports $\sum_i d_i$ with approximate standard error $\sqrt{\sum_g n_g s_{d,g}^2}$, respecting fixed group counts, treating records as independent and holding the trained fits fixed. These errors do not include training-set uncertainty, repeat-runner dependence or variation across races.

Reference	Mixture minus skew-normal (nats)	Standard error
25th percentile (primary)	244.9	48.1
None (raw log ratio)	108.6	42.7
10th percentile	194.6	46.4
50th percentile	280.8	47.8

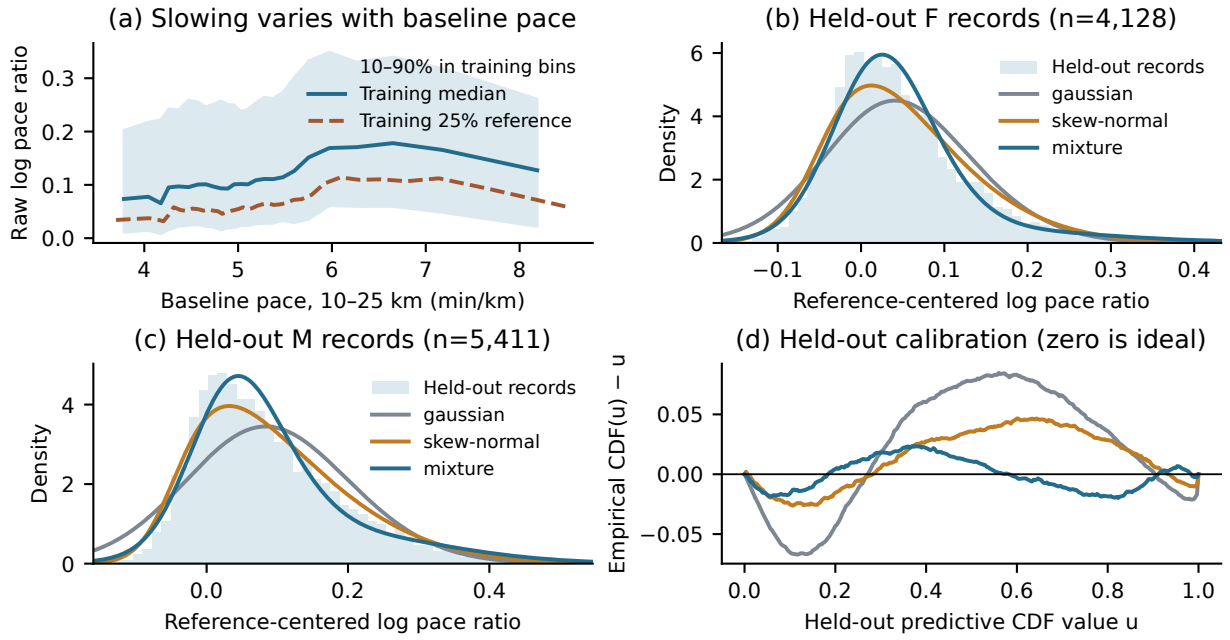


Figure 1: (a) Training quantiles; (b–c) held-out histograms and fitted densities, zoomed to the bulk; (d) predictive-CDF calibration, where zero is ideal. Scores include all tails; the mixture still shows misfit.

The primary gain is 0.0257 nats per record over the skew-normal, and 1151.5 total nats (s.e. 72.7) over the Gaussian. Using each group's training 5th/95th percentiles to fix the regions, the central interval contributes -28.0 nats (-11.4% of the net gain), the upper tail 196.0 (80.0%), and the lower tail 76.9 (31.4%); signed contributions can cancel. All 4 primary mixture densities have one mode. Their component SD ratios $\hat{\sigma}_{1,g}/\hat{\sigma}_{0,g}$ range 2.117–2.463 (variance ratios 4.481–6.065). The wider higher-mean component has responsibility $> 1/2$ in both tails: its left crossover $y^*_{\text{left},g}$ ranges from -0.159 to -0.135, below which it dominates despite negative centered y. Membership thus does not identify directional slowing or physiological wall states. A skew-normal is only one possible unimodal alternative.

(6) Next step. In a new, uninterrupted race including nonfinishers, I would compare segment trajectories with independently recorded exhaustion onset, accounting for course location and initial pacing. This tests whether abrupt, sustained slowing coincides with exhaustion. The current split remains an internal diagnostic: earlier work explored these data, and all sensitivities reuse the holdout.